PHB_221_Final_Project

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Main Questions

Main Question: How does the age-adjusted death rate for heart disease vary across states in the United States, and are there any observable regional or geographic trends over time?

- Research Objectives:
  - Compare state-level heart disease mortality rates
  - Identify patterns
  - Explore how rates change over time
  - Determine whether areas with consistently higher mortality are geographically concentrated
  - Can past trends in age adjusted heart disease death rates be used to predict future state level rates?

Dataset and Libraries

```r
# libraries
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3

## Warning: package 'purrr' was built under R version 4.3.3

## Warning: package 'lubridate' was built under R version 4.3.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.3.3

## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
```

```
##     expand, pack, unpack
```

```r
library(Metrics)
```

```
## Warning: package 'Metrics' was built under R version 4.3.3
```

```r
# dataset
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv")
```

```
## Rows: 10868 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): 113 Cause Name, Cause Name, State
## dbl (3): Year, Deaths, Age-adjusted Death Rate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Data Collection and Cleaning

```r
# view first few rows of data
head(data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Accidents (unintention~ Unintention~ Unit~ 169936                   49.4
## 2  2017 Accidents (unintention~ Unintention~ Alab~   2703                   53.8
## 3  2017 Accidents (unintention~ Unintention~ Alas~    436                   63.7
## 4  2017 Accidents (unintention~ Unintention~ Ariz~   4184                   56.2
## 5  2017 Accidents (unintention~ Unintention~ Arka~   1625                   51.8
## 6  2017 Accidents (unintention~ Unintention~ Cali~  13840                   33.2
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
# check the number of rows and cols
dim(data)
```

```
## [1] 10868     6
```

```r
# filter data
filtered_data <- data[data$`Cause Name` == "Heart disease", ]

head(filtered_data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Diseases of heart (I00~ Heart disea~ Unit~ 647457                    165
## 2  2017 Diseases of heart (I00~ Heart disea~ Alab~  13110                    223.
## 3  2017 Diseases of heart (I00~ Heart disea~ Alas~    814                    135
## 4  2017 Diseases of heart (I00~ Heart disea~ Ariz~  12398                    142.
## 5  2017 Diseases of heart (I00~ Heart disea~ Arka~   8270                    224.
## 6  2017 Diseases of heart (I00~ Heart disea~ Cali~  62797                    143.
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
dim(filtered_data)
```

```
## [1] 988   6
```

```r
# missing data
colSums(is.na(filtered_data))
```

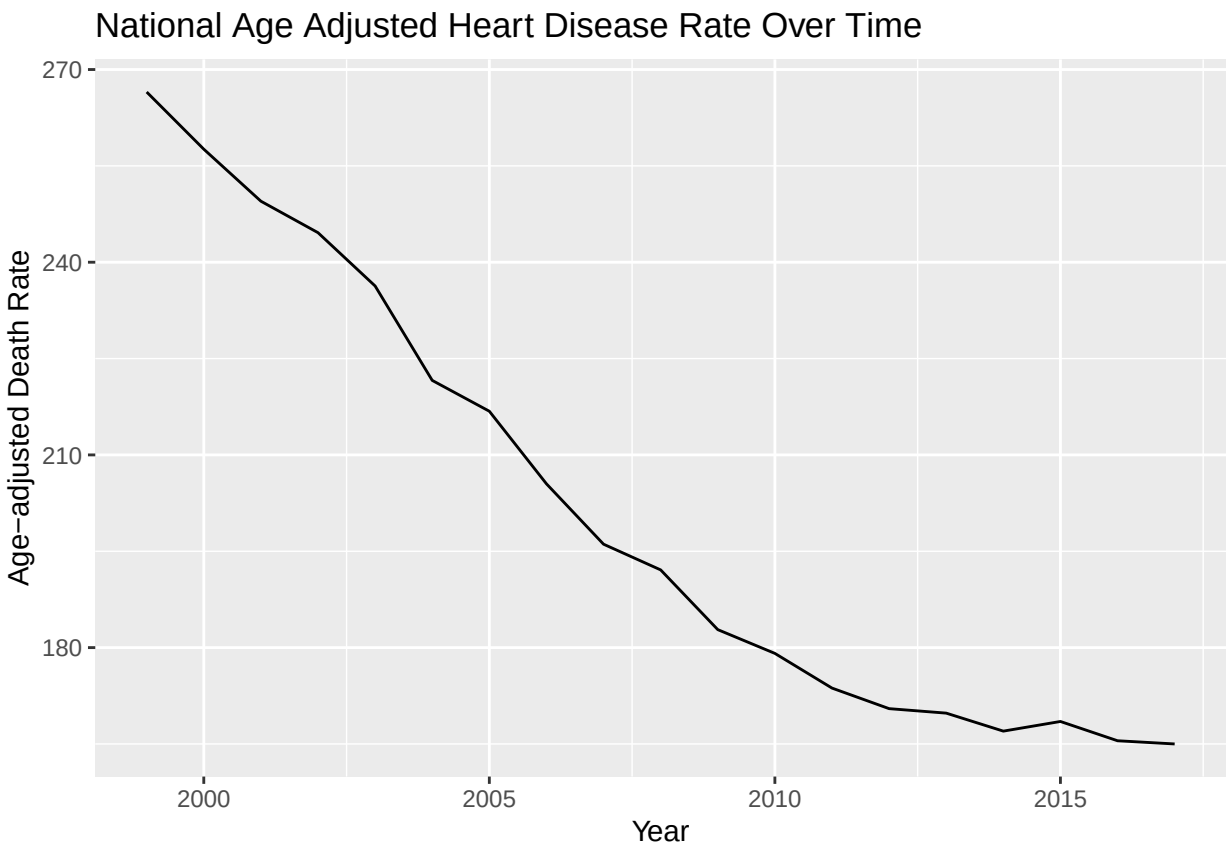
```
##                     Year          113 Cause Name              Cause Name
##                        0                       0                       0
##                    State                  Deaths Age-adjusted Death Rate
##                        0                       0                       0
```

Explantory Analysis

```r
# national level
national_data <- data |>
  filter(`Cause Name` == "Heart disease", State == "United States")

head(national_data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Diseases of heart (I00~ Heart disea~ Unit~ 647457                   165
## 2  2016 Diseases of heart (I00~ Heart disea~ Unit~ 635260                   166.
## 3  2015 Diseases of heart (I00~ Heart disea~ Unit~ 633842                   168.
## 4  2014 Diseases of heart (I00~ Heart disea~ Unit~ 614348                   167
## 5  2013 Diseases of heart (I00~ Heart disea~ Unit~ 611105                   170.
## 6  2012 Diseases of heart (I00~ Heart disea~ Unit~ 599711                   170.
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
ggplot(data = national_data, mapping = aes(x = Year, y = `Age-adjusted Death Rate`)) +
  geom_line() +
  labs(title = "National Age Adjusted Heart Disease Rate Over Time")
```

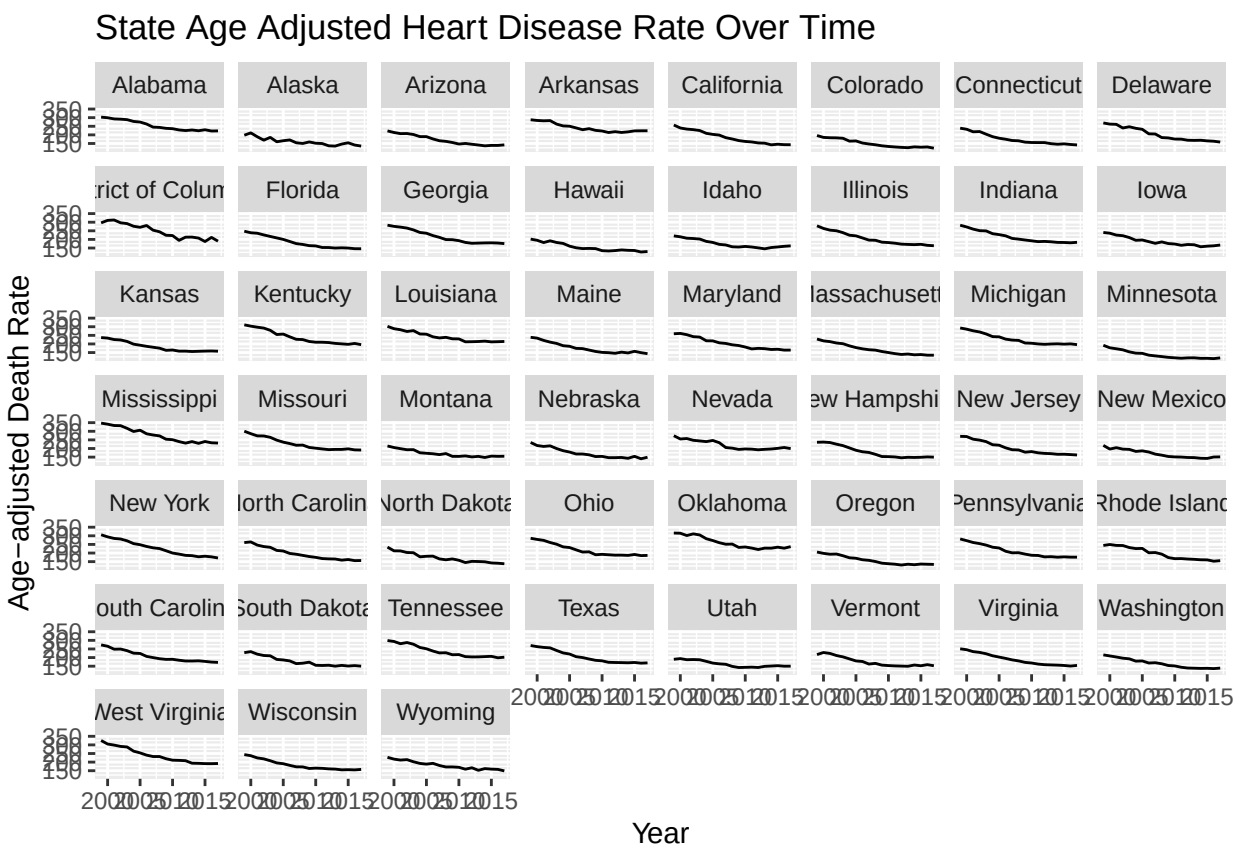



```r
# state level

state_only <- filtered_data |>
  filter(State != "United States") |>
  mutate(
    Region = case_when(
      State %in% c("Maine", "New Hampshire", "Vermont", "Massachusetts",
                   "Rhode Island", "Connecticut", "New York", "New Jersey",
                   "Pennsylvania") ~ "Northeast",
      State %in% c("Ohio", "Indiana", "Illinois", "Michigan", "Wisconsin",
                   "Minnesota", "Iowa", "Missouri", "North Dakota", "South Dakota",
                   "Nebraska", "Kansas") ~ "Midwest",
      State %in% c("Delaware", "Maryland", "District of Columbia", "Virginia",
                   "West Virginia", "North Carolina", "South Carolina", "Georgia",
                   "Florida", "Kentucky", "Tennessee", "Alabama", "Mississippi",
                   "Arkansas", "Louisiana", "Oklahoma", "Texas") ~ "South",
      State %in% c("Montana", "Idaho", "Wyoming", "Colorado", "New Mexico",
                   "Arizona", "Utah", "Nevada", "Washington", "Oregon", "California",
                   "Alaska", "Hawaii") ~ "West"
    )
  )


head(state_only)
```

```
## # A tibble: 6 x 7
##    Year `113 Cause Name` `Cause Name` State Deaths Age-adjusted Death R~1 Region
```

```
##   <dbl> <chr>            <chr>        <chr>  <dbl>                    <dbl> <chr>
## 1  2017 Diseases of hea~ Heart disea~ Alab~  13110                     223. South
## 2  2017 Diseases of hea~ Heart disea~ Alas~    814                     135  West
## 3  2017 Diseases of hea~ Heart disea~ Ariz~  12398                     142. West
## 4  2017 Diseases of hea~ Heart disea~ Arka~   8270                     224. South
## 5  2017 Diseases of hea~ Heart disea~ Cali~  62797                     143. West
## 6  2017 Diseases of hea~ Heart disea~ Colo~   7060                     123. West
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

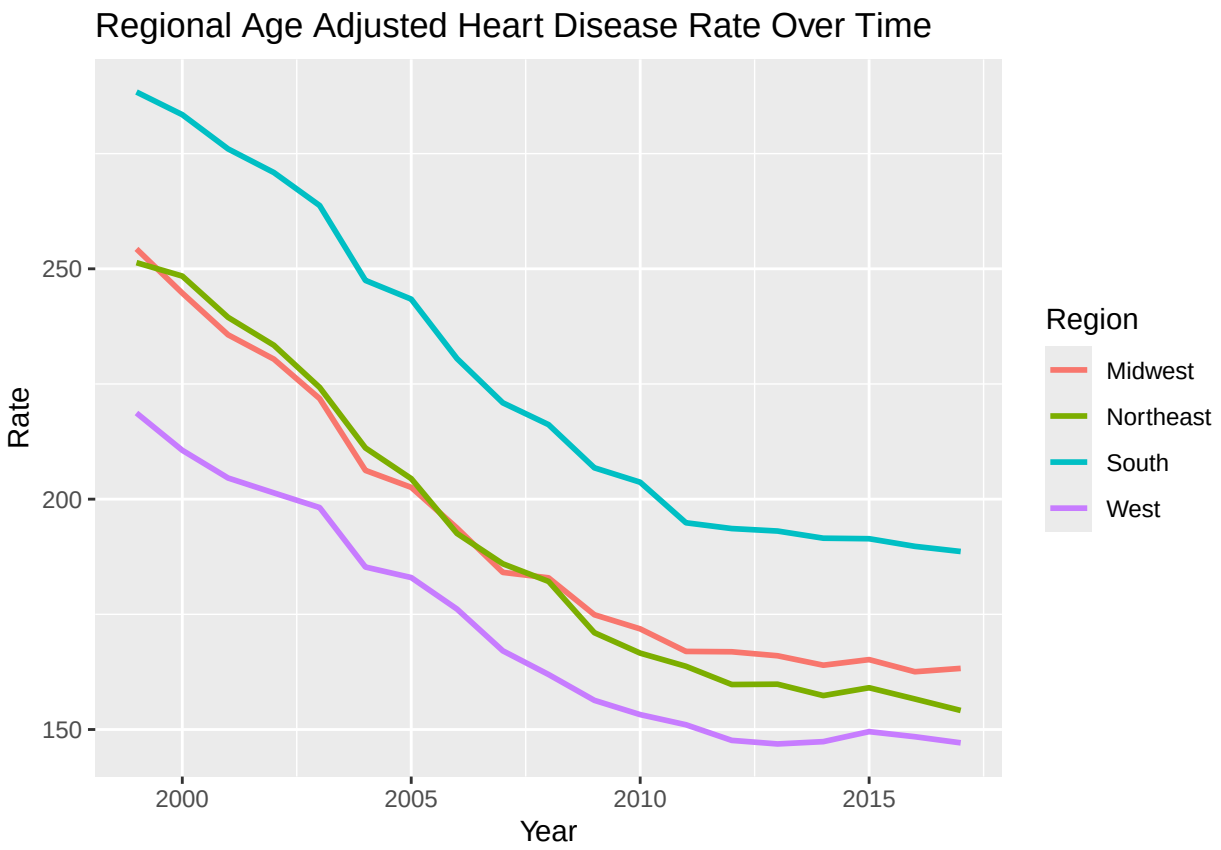
```
ggplot(data = state_only, mapping = aes(x = Year, y = `Age-adjusted Death Rate`)) +
  geom_line() +
  facet_wrap(~ State) +
  labs(title = "State Age Adjusted Heart Disease Rate Over Time")
```



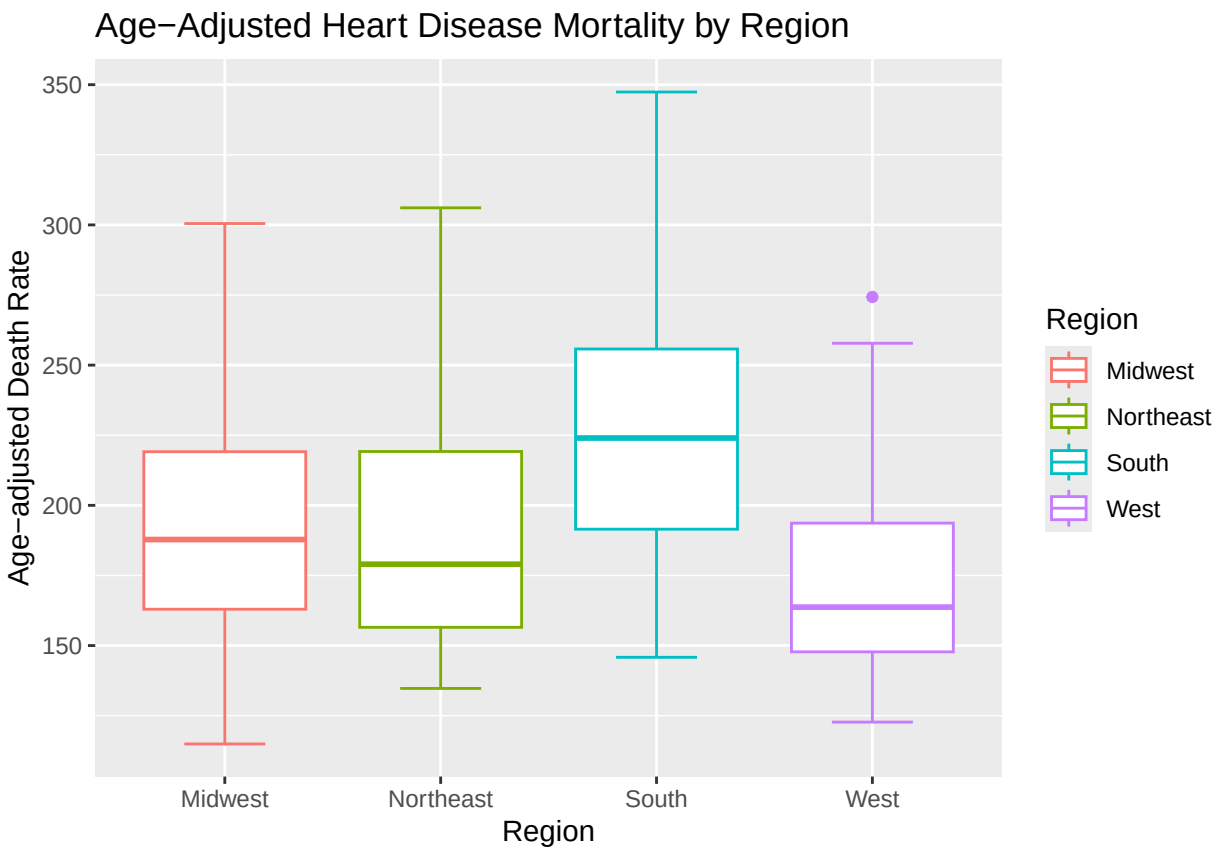

```
# regional level
regional_summary <- state_only |>
  group_by(Region, Year) |>
  summarise(
    Rate = mean(`Age-adjusted Death Rate`)
  )
```

```
## `summarise()` has grouped output by 'Region'. You can override using the
## `.groups` argument.
```

```
ggplot(data = regional_summary, mapping = aes(x = Year, y = Rate, color = Region)) +
  geom_line(linewidth = 1) +
  labs(title = "Regional Age Adjusted Heart Disease Rate Over Time")
```




```
ggplot(data = state_only, mapping = aes(x = Region, y = `Age-adjusted Death Rate`, color = Region)) +
  geom_boxplot(staplewidth = 0.5) +
  labs(title = "Age-Adjusted Heart Disease Mortality by Region")
```



Statistical Testing

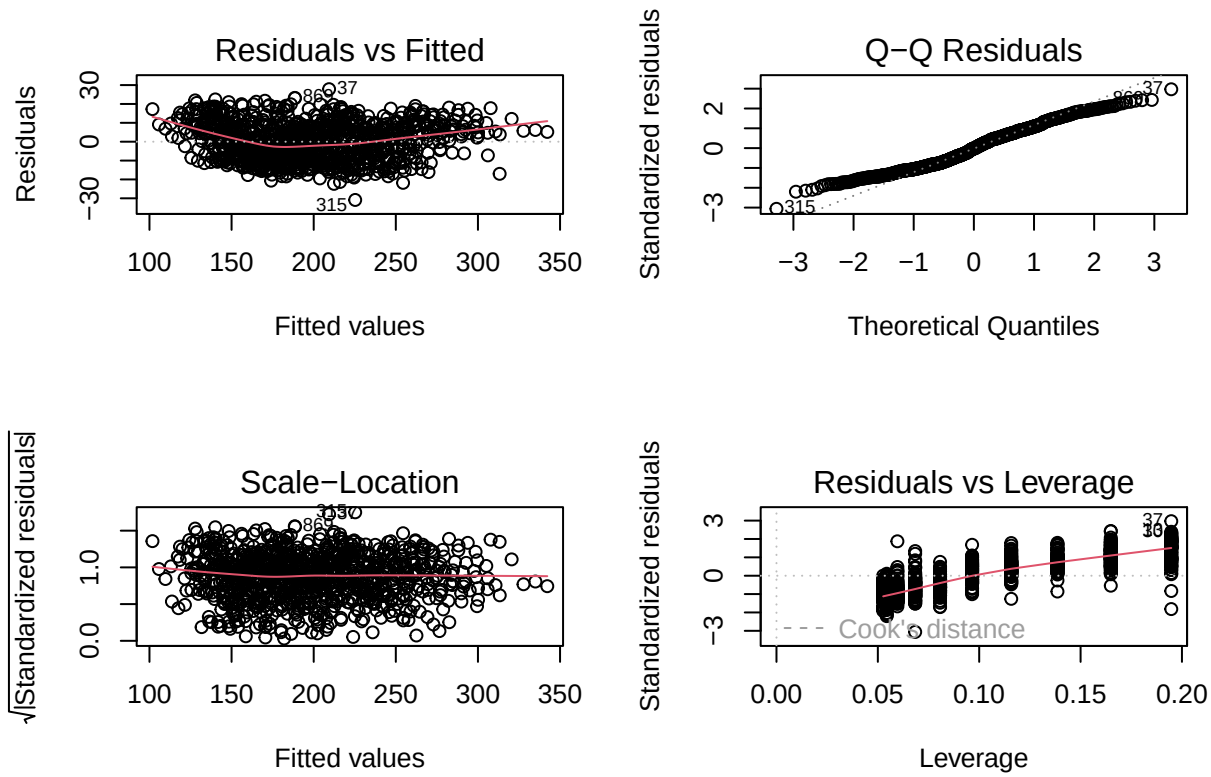
- How does mortality vary across states, and are there regional/geographical patterns over time?
- one way anova limitations:
  - has to be done using a single year, not all years
  - anova assumes independent groups
  - using all years violate independence
- two way anova
- notes:
  - https://www.geeksforgeeks.org/data-science/analysis-of-variance-anova/
  - cook's distance
    - How does a single observation (data pt) influece the entire model?
    - If a data pt is removed, does the model's prediction change a lot?

```
# two way anova testing
# * (interaction term) bc each state has their own slope, if + was used then slope for Year is the same
two_way <- aov(`Age-adjusted Death Rate` ~ State * Year, data = state_only)
summary(two_way)
```

```
##              Df Sum Sq Mean Sq  F value Pr(>F)
## State        50 957770   19155  175.520 <2e-16 ***
## Year          1 842449  842449 7719.314 <2e-16 ***
## State:Year   50  37132     743    6.805 <2e-16 ***
## Residuals   867  94620     109
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# check assumptions: homoscedasticity, normalit of residuals, linearity, and influential pts
par(mfrow = c(2,2))
plot(two_way)
```




```r
D <- cooks.distance(two_way)
sum(D > 1)
```

```
## [1] 0
```

Predictive Modeling

- Can past trends in age adjusted heart disease death rates be used to predict future state level rates?
- notes:
  - simple linear regression wont work bc we need year and state
    - assumes all states share the same mean and trend over time
  - multiple linear regression year + state
    - treats state as a categorical predictor (fixed state effects)
    - each state getes it's own intercept (baseline level)
    - BUT it assumes same slope for all states
  - multiple linear regression year * state
    - different slope per state

```r
# simple linear regression
m1 <- lm(`Age-adjusted Death Rate` ~ Year, data = state_only)
#summary(m1)
```

```r
# multiple regression +
m2 <- m_additive <- lm(`Age-adjusted Death Rate` ~ Year + State, data = state_only)
#summary(m2)

# multiple regression *
m3 <- lm(`Age-adjusted Death Rate` ~ Year * State, data = state_only)
#summary(m3)

# mixed model?

mixed1 <- lmer(`Age-adjusted Death Rate` ~ Year + (1 | State), data = state_only)
#summary(mixed1)

mixed2 <- lmer(`Age-adjusted Death Rate` ~ Year + (Year | State), data = state_only)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## unable to evaluate scaled gradient
```

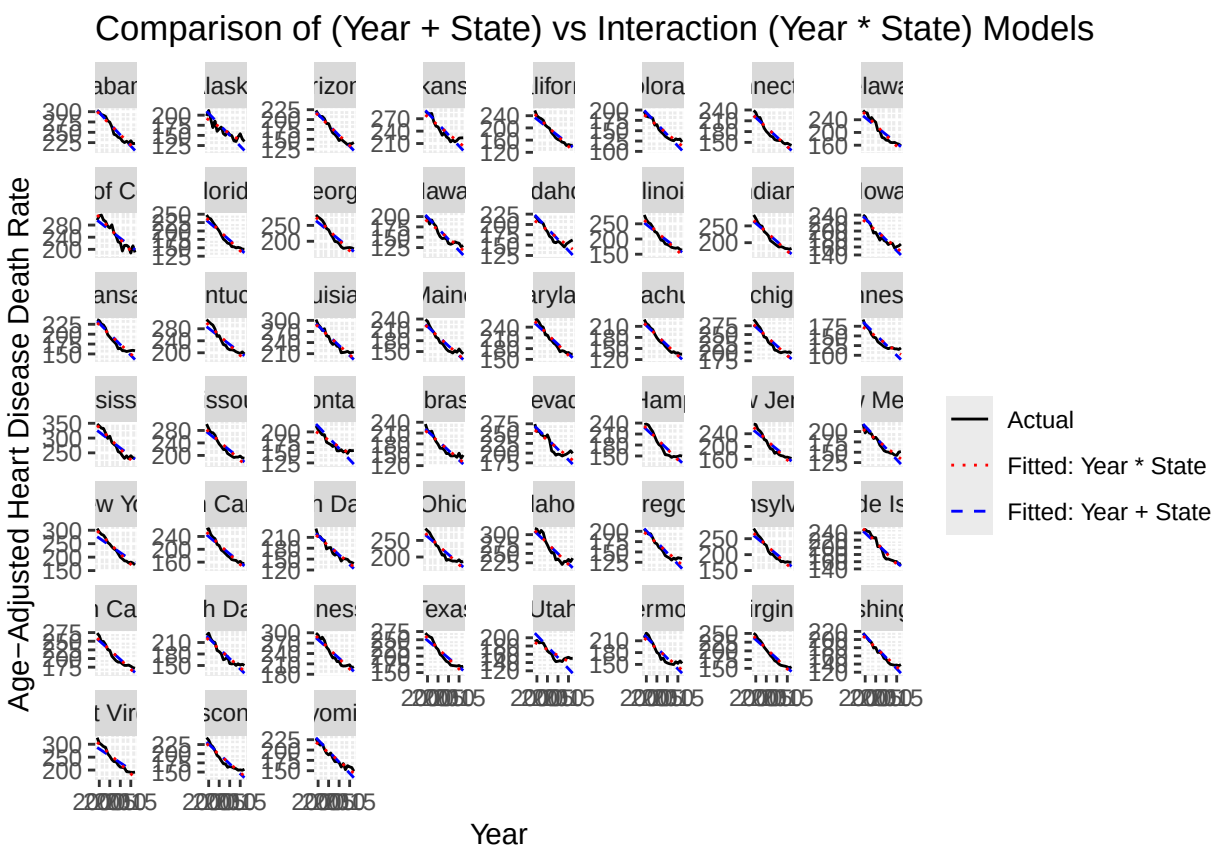
```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge: degenerate Hessian with 2 negative eigenvalues
```

```r
#summary(mixed2)


# plot comparison

state_only <- state_only |>
  mutate(
    fitted_m1 = fitted(m1),
    fitted_m2 = fitted(m2),
    fitted_m3 = fitted(m3),
    fitted_mixed1 = fitted(mixed1),
    fitted_mixed2 = fitted(mixed2)
  )

head(state_only)
```

```
## # A tibble: 6 x 12
##    Year `113 Cause Name` `Cause Name` State Deaths Age-adjusted Death R~1 Region
##   <dbl> <chr>            <chr>        <chr>  <dbl>                  <dbl> <chr>
## 1  2017 Diseases of hea~ Heart disea~ Alab~  13110                   223. South
## 2  2017 Diseases of hea~ Heart disea~ Alas~    814                   135  West
## 3  2017 Diseases of hea~ Heart disea~ Ariz~  12398                   142. West
## 4  2017 Diseases of hea~ Heart disea~ Arka~   8270                   224. South
## 5  2017 Diseases of hea~ Heart disea~ Cali~  62797                   143. West
## 6  2017 Diseases of hea~ Heart disea~ Colo~   7060                   123. West
## # i abbreviated name: 1: `Age-adjusted Death Rate`
## # i 5 more variables: fitted_m1 <dbl>, fitted_m2 <dbl>, fitted_m3 <dbl>,
## #   fitted_mixed1 <dbl>, fitted_mixed2 <dbl>
```

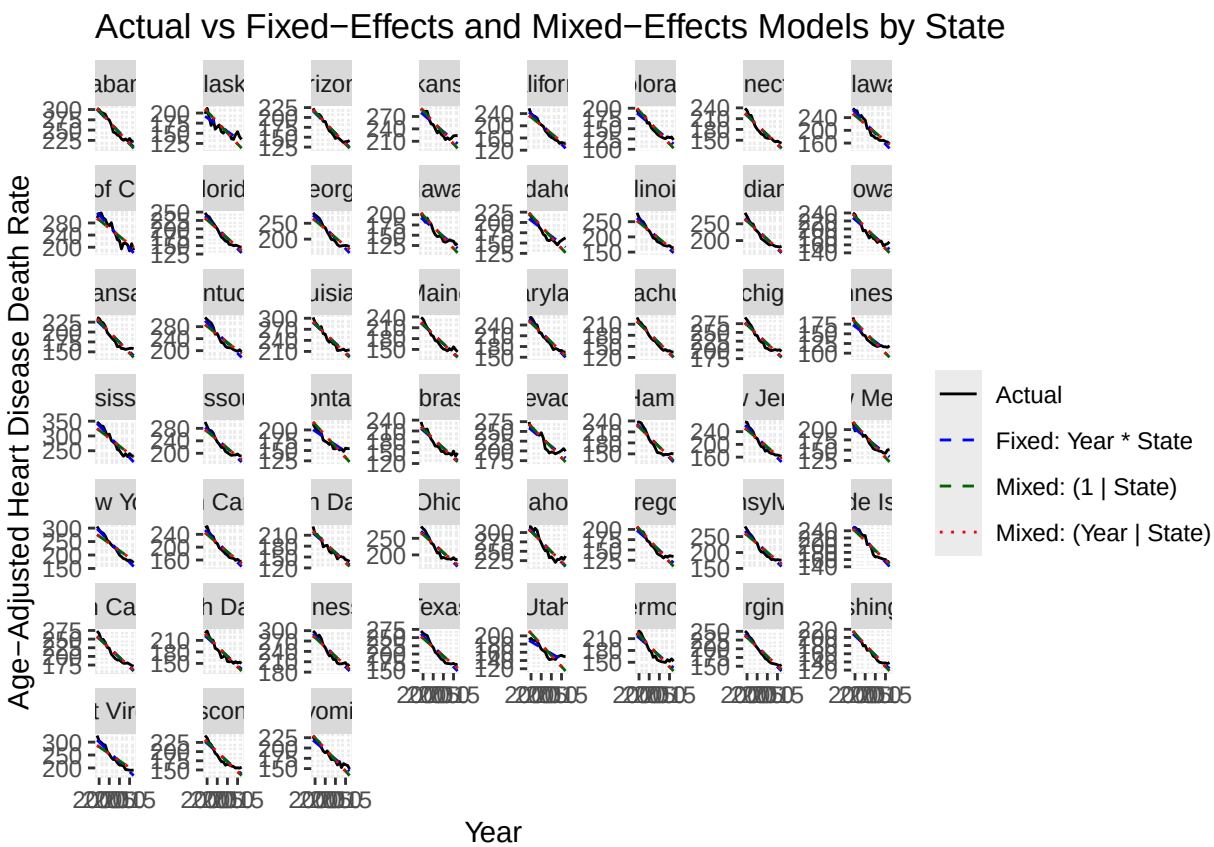
```r
ggplot(data = state_only, mapping = aes(x = Year)) +

  # actual data
  geom_line(mapping = aes(y = `Age-adjusted Death Rate`, color = "Actual")) +
```

```r
  # year + state
  geom_line(mapping = aes(y = fitted_m2, color = "Fitted: Year + State"), linetype = "dashed") +

  # year * state
  geom_line(mapping = aes(y = fitted_m3, color = "Fitted: Year * State"), linetype = "dotted") +

  facet_wrap(~ State, scales = "free_y") +

  labs(
    title = "Comparison of (Year + State) vs Interaction (Year * State) Models",
    y = "Age-Adjusted Heart Disease Death Rate",
    color = ""
  ) +

  scale_color_manual(values = c(
    "Actual" = "black",
    "Fitted: Year + State" = "blue",
    "Fitted: Year * State" = "red"
  ))
```




```r
ggplot(data = state_only, mapping = aes(x = Year)) +
  # actual data
  geom_line(mapping = aes(y = `Age-adjusted Death Rate`, color = "Actual")) +

  # m3
  geom_line(aes(y = fitted_m3, color = "Fixed: Year * State"), linetype = "dashed") +
```

```r
  # mixed1
  geom_line(aes(y = fitted_mixed1, color = "Mixed: (1 | State)"), linetype = "dashed") +

  # mixed2
  geom_line(mapping = aes(y = fitted_mixed2, color = "Mixed: (Year | State)"), linetype = "dotted") +

  facet_wrap(~ State, scales = "free_y") +

  labs(
    title = "Actual vs Fixed-Effects and Mixed-Effects Models by State",
    y = "Age-Adjusted Heart Disease Death Rate",
    color = ""
  ) +

  scale_color_manual(values = c(
    "Actual"               = "black",
    "Fixed: Year * State" = "blue",
    "Mixed: (1 | State)"  = "darkgreen",
    "Mixed: (Year | State)" = "red"
  ))
```



- Train Models:
  - 80/10/10 does not work bc we cant randomly shuffle years

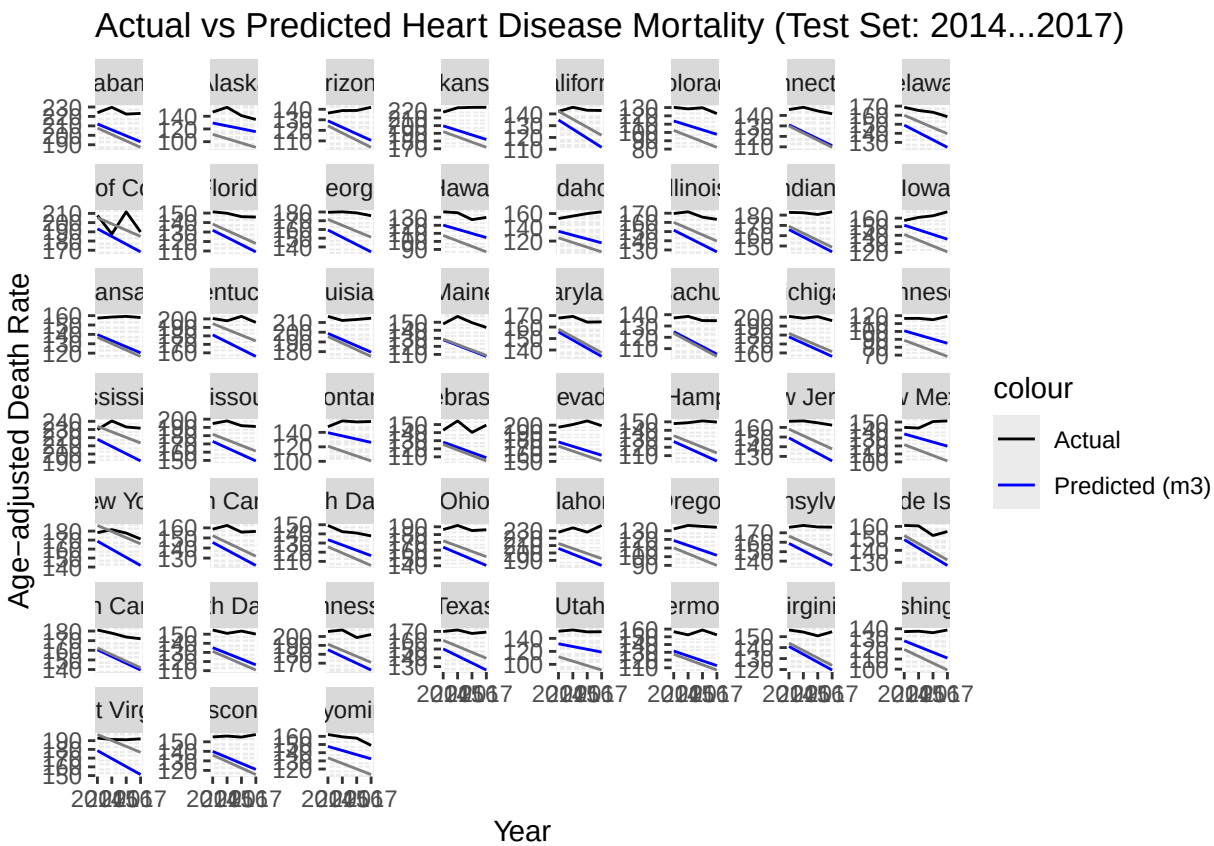
```r
train <- state_only |>
  filter(Year <= 2013)

test <- state_only |>
```

```r
  filter(Year > 2013)

# train models
m2_train <- lm(`Age-adjusted Death Rate` ~ Year + State, data = train)
m3_train <- lm(`Age-adjusted Death Rate` ~ Year * State, data = train)
mixed1_train <- lmer(`Age-adjusted Death Rate` ~ Year + (1 | State), data = train)
mixed2_train <- lmer(`Age-adjusted Death Rate` ~ Year + (Year | State), data = train)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
# predict on test set
test$pred_m2 <- predict(m2_train, newdata = test)
test$pred_m3 <- predict(m3_train, newdata = test)
test$pred_mixed1 <- predict(mixed1_train, newdata = test)
test$pred_mixed2 <- predict(mixed2_train, newdata = test)

ggplot(data = test, mapping = aes(x = Year)) +

  geom_line(mapping = aes(y = `Age-adjusted Death Rate`, color = "Actual")) +

  geom_line(mapping = aes(y = pred_m3, color = "Predicted (m3)")) +

  geom_line(mapping = aes(y = pred_mixed1, color = "Predicted (m1)")) +

  facet_wrap(~ State, scales = "free_y") +

  labs(title = "Actual vs Predicted Heart Disease Mortality (Test Set: 2014-2017)") +

  scale_color_manual(values = c("Actual" = "black", "Predicted (m3)" = "blue", "Predicted (mixed m1)" =
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <93>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <93>
```



Evaluation and Interpretation

- look at R^2, p vals, residuals

```
# check models accuracies
# RMSE and MAE
rmse_m2 <- rmse(test$`Age-adjusted Death Rate`, test$pred_m2)
mae_m2  <- mae(test$`Age-adjusted Death Rate`, test$pred_m2)

rmse_m3 <- rmse(test$`Age-adjusted Death Rate`, test$pred_m3)
mae_m3  <- mae(test$`Age-adjusted Death Rate`, test$pred_m3)

rmse_mixed1 <- rmse(test$`Age-adjusted Death Rate`, test$pred_mixed1)
mae_mixed1  <- mae(test$`Age-adjusted Death Rate`, test$pred_mixed1)

rmse_mixed2 <- rmse(test$`Age-adjusted Death Rate`, test$pred_mixed2)
mae_mixed2  <- mae(test$`Age-adjusted Death Rate`, test$pred_mixed2)

# accuracies
data.frame(
  Model = c("m2 (Year+State)", "m3 (Year*State)",
            "mixed1 (1|State)", "mixed2 (Year|State)"),
  RMSE = c(rmse_m2, rmse_m3, rmse_mixed1, rmse_mixed2),
  MAE  = c(mae_m2, mae_m3, mae_mixed1, mae_mixed2)
)
```

```
##                   Model      RMSE       MAE
```

```
## 1     m2 (Year+State) 28.99761 26.20274
## 2     m3 (Year*State) 27.51421 25.92697
## 3    mixed1 (1|State) 28.96682 26.19361
## 4 mixed2 (Year|State) 28.98851 26.19928
```

Between the two models, the predictions follow a smooth linear decline because the models learn long term state specific trends but it can't capture short term year to year flucatuatons. Based on the RMSE and MAE Year x State (m3) performed the best as it showed the lowest scores, whicbh means allowing each state to have its own slope improves predictive model performance. Mixed effects were a bit less accurate because it tries to make the slopes to be more similar across states.